

Alan Luna

Trust and Transparency in the Digital Age

I. Introduction

Modern organizations, including technology companies and government entities, rely heavily on digital platforms to deliver services, collect information, and support decision making. However, many organizations face engagement in these systems due to long-standing, invasive privacy practices that have fostered user mistrust and heightened privacy concerns. Currently, 79% of Americans are concerned over how companies use their data (Pew Research Center, 2019). Over time, the practices have weakened users' confidence in how their personal data is collected, monitored and shared, creating a critical barrier to effective digital user-interaction.

Public discourse and empirical research indicate that users are increasingly aware of and sensitive to how their personal information is being collected and used. Pew Research Center indicated that as of 2023, 70% of people in a study reported feeling little control over how companies use their data (Pew Research Center, 2023). Concerns surrounding surveillance, data misuse and loss of personal control have also intensified and increased as digital systems become more pervasive and less transparent (Acquisti, Brandimarte and Loewenstein, 2015). While data-driven systems promise efficiency and innovation they also introduce psychological and behavioral consequences that can influence how individuals engage with digital platforms. Table 1 summarizes the conceptual relationship between organizational data collection practices, user psychological responses and their outcomes.

Research in psychology and communication studies also demonstrates that perceptions of privacy and data surveillance are associated with heightened cognitive load, emotional distress, and overall reduced

willingness to engage with digital technologies (Buchi, Fosch-Villaronga and Tamo-Larrieux, 2021). Users who feel monitored or inadequately informed about data practices often respond by limiting platform use, withholding information and avoiding any digital systems all together (Diven and Hart, 2006). These responses reflect broader issues of trust, perceived control and psychological safety factors that play a critical role in shaping an individual's digital behavior.

From the perspective of the organizations, the users' responses to privacy practices have important implications. Digital platforms are most effective when users actively participate, provide accurate information and adopt new features or systems when they are introduced. When trust is weakened, organizations may experience lower engagement rates as well as reduced data quality while creating friction and possibly distress in users.

II. Business Intelligence Methods

a. The Business Process

The project follows a structured Business Intelligence process designed to transform complex survey data into actionable insight that support an understanding of digital trust, privacy concerns and engagement from the perspective of the user and an organization. The analytical process focused on how user perceptions of privacy and control shape engagement with digital platforms, with emphasis on organizational decision-making and ethical data practices. Each stage is intentionally sequenced in a way that ensures that the analytical outputs remained aligned with organizational decision-making needs rather than isolated statistical results.

Table 1. Conceptual Pathway from Data Practices to Digital Engagement		
Organizational Data Practices	User Psychological Response	Digital Platform Outcomes
Data collection methods Privacy policy design Transparency level	Perceived control Privacy concern Trust	Platform engagement System adoption Willingness to share data

At the outset, the analytical focus was refined to ensure that the project addressed both user psychology and the performance of organizations. Rather than examining attitudes in isolation, the process looked at how such attitudes could translate into behavioral outcomes that affect the effectiveness of digital systems. Following the framing of the problem, attention then shifted to whether available data could meaningfully support the analytical objectives. The dataset used in this project contains a large, nationally representative sample of U.S. adults and includes multiple indicators of trust, perceived control, and privacy concerns. While the raw data was broad and provided strong analytical potential, it did require substantial transformation to be useful in the context of the project.

The business process then moved into preparation and analysis. Visualizations and correlations were done as a means of exploratory analysis to explore the relationships between variables, additionally, these activities were done under the context of understanding the underlying psychological constructs so that the analytical methods would yield interpretable results. Once completed, the data was analyzed using a combination of descriptive, inferential, and validation techniques to allow patterns to be identified and tested for stability.

Regarding anticipated outcomes, the process is expected to yield several meaningful outcomes for organizations that operate digital platforms. First, it provides a clear understanding of how privacy related perceptions influence trust and engagement in the organization and their digital platforms. Second, it identifies which attitudes are most strongly associated with reduced participation or resistance to new systems. Third, it also generates evidence-based insights that can inform decision about transparency, communication strategies, platform design, privacy policy practices and overall digital interaction with users. Collectively, these outcomes support organizational productivity by reducing uncertainty and increasing user participation while supporting ethical business practices relating to user data collection.

b. Data and Data Structures

The data used for this project come from the Pew Research Center's American Trends Panel, Wave 127, which includes responses from approximately 5,100 U.S. adults and is designed to be nationally representative. The survey was administered online and is based on questions that contains a combination of ordinal, nominal, and binary variables, all related to technology use, privacy attitudes, trust in institutions and demographic characteristics. The breadth of the dataset makes it well suited to examine relationships between psychological perceptions and digital behaviors. However, the data was distributed using a lot of different questions asking about concepts but not direct psychological constructs and required conceptual alignment before analysis could begin. Addressing these challenges was a central component of the process before beginning analyses.

Several steps were followed to ensure that the data could meaningfully support the analysis. Variables were renamed and reorganized to improve interpretability and consistency across analysis. Second, the dataset file is an sav. file with labeled values the 'as.numeric()' function applied to convert variables into numeric values. Non-informative responses such as "Don't Know" or "Refused" were originally coded as values such as "99" or "9", these responses were recoded as missing values using base R logical indexing to prevent artificial inflation of the summary statistics and to ensure the analysis reflected the true user responses. Cases with missing values were then removed using complete case filtering, then creating a cleaned dataset.

c. Statistical Methodology

All statistical methodology was conducted in R using a combination of base R, the 'psych', 'ggplot2' and 'caret' packages. The analytical strategy progressed from descriptive and exploratory analyses to probability-based reasoning, hypothesis testing via regression and group comparisons and validation using k-fold cross validation.

A key part of the project was the construction of two composite scores, digital mistrust and digital privacy concerns. The composite scores were developed by combining multiple Likert-scale items that ask about the

concepts related to their respective constructs such as trust in organization leaders, organizational data usage, perceived risk, concern over data practices and perceived control. Items were selected based on conceptual relevance and prior empirical research and were reverse coded where necessary to ensure directional consistency and standardized using z-scoring using the ‘scale()’ function. The items were then combined by computing the row-wise mean of standardized items using ‘rowMeans()’. Internal consistency and construct validity were assessed using Cronbach’s alpha and exploratory factor analysis via the ‘psych’ package in R. Parallel analysis informed factor retention decisions.

Analyses began with descriptive statistics to establish baseline distributions and assess the structure of the key variables. Central tendency and dispersion were examined using base R summary functions, ‘summary()’, applied to standardized and non-standardized composite scores as well as the policy related indicators such as perceived policy effectiveness ‘ppeff’ and privacy literacy ‘priv_lit’. These summaries provided initial insight into the prevalence of digital mistrust and privacy concerns in the sample.

Exploratory visualizations were created using ‘ggplot2’ to assess distributions and visualize skewness and evaluate the potential relationships prior to modeling. Histograms, density overlays, scatterplots with linear trend lines and boxplots by demographic groups were used to assess normality, linearity and group differences

Applied probability was used to understand statistical measures into quantities. Because ‘trust_z’, the standardized mistrust composite score, is standardized, values above zero were interpreted as above average digital mistrust. A binary indicator ‘high_mistrust’ was created using ‘ifelse()’ functions to classify respondents as high or low mistrust. Marginal, joint and conditional probabilities were then computed using sample proportions such as: “The probability of a random adult exhibits high digital mistrust”, “The probability of high mistrust conditional on smartphone ownership” and “Conditional probabilities of high mistrust given privacy literacy, digital tracking behaviors and elevated privacy concerns”. These probability calculations provided interpretable estimates of association that support the business intelligence reasoning prior to conducting formal hypothesis testing.

When assessing linear associations among the key variables in the project, a correlation matrix was computed after grouping the key variables in a data matrix and using the function ‘cor()’. This analysis provided an early assessment of direction and strengths of the relationships between digital mistrust, privacy concerns, privacy literacy, digital habits and demographic characteristics which helped justify use certain predictors for later regression models.

Table 2. Hypothesis Testing Plan				
#	Null Hypothesis	Alt. Hypothesis	H ₀ (symbolic)	H _a (symbolic)
1	Privacy concerns do not predict digital mistrust	Higher privacy concerns predict higher digital mistrust	$\beta_{conc} = 0$	$\beta_{conc} > 0$
2	Privacy literacy does not predict digital mistrust.	Greater privacy literacy predicts lower digital mistrust.	$\beta_{priv_lit} = 0$	$\beta_{priv_lit} < 0$
3	Perceived privacy policy effectiveness does not predict digital mistrust.	Higher perceived policy effectiveness predicts lower digital mistrust.	$\beta_{ppeff} = 0$	$\beta_{ppeff} < 0$
4	Digital habits have no joint effect on mistrust	Digital habits have a joint effect on mistrust.	$\beta_{track} = \beta_{encrypt} = \beta_{browser} = \beta_{social} = 0$	At least one $\beta \neq 0$
5	Digital mistrust does not differ by age group	Digital mistrust differs by age group.	$\mu_{18-29} = \mu_{30-49} = \mu_{50-64} = \mu_{65+}$	At least one group mean differs

Formal hypothesis testing was also conducted using regression models and group comparisons. Regression coefficients, F-tests and confidence intervals and likelihood-based measures served as inferential tests of the

hypothesis. Additionally, each hypothesis was tested using two tailed tests with $\alpha = .05$. The variables included within the analysis are Privacy Concerns composite (conc), Digital Mistrust composite (trust), perceived privacy policy effectiveness (ppeff), privacy literacy (priv_lit), digital habits (tracking decline, encryption in messages, private browser, social media settings) and age (Table 2.).

Multiple linear regression was used to model digital mistrust as a function of privacy concerns, privacy literacy, perceived policy effectiveness, digital habits, and demographic variables. The models were estimated using 'lm()' and individual coefficient significance was evaluated using t-test from 'summary()' while joint significance was assessed using 'anova()'. Additionally, to see if digital mistrust and privacy concerns varied across demographic groups, ANOVA models were used with 'aov()'. Separate models examined differences by age categories, educational level, and gender. Because several of the questions involved binary outcomes, logistic regression was also used to model the probability of high digital mistrust. The binary outcome was created earlier for applied probability and recoded as a factor for modeling. The logistic models were created using 'glm()'.

Additionally, to assess model stability, 10 fold cross-validation was used to the linear regression model using the 'caret' package. The 'train()' function was used repeatedly to estimate the model across multiple data partitions. Cross-validation provided a more realistic estimate to how well the model would generalize new data with strengthened its use in BI applications.

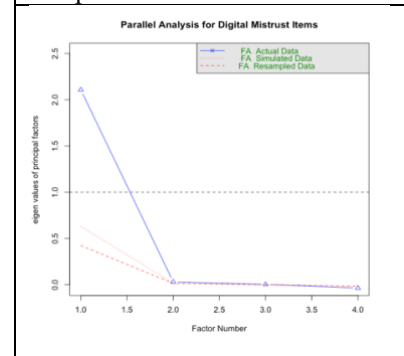
Some alternative approaches were also considered but due to findings and challenges in the data were not adopted. Since the Perceived Control composite was found to load less well on a single factor as compared to the mistrust factor, models based on that score were not approached. Initially, the project also focused on solely linear regression modeling, but after analyzing applied probability and hypothesis testing it was found that a binary outcome was also appropriate for the project.

III. Results

To assess the quality and appropriateness of these composite measures, reliability and validation testing were conducted using the 'psych' package in R including 'psych::alpha' and 'fa()'. Internal consistency was evaluated using Cronbach's alpha using with thresholds of $\alpha \geq .70$ considered acceptable, $\alpha \geq .80$ considered good and $\alpha \geq .90$ considered excellent reliability (Altmann and Allen, 2021; Takavol and Dennick, 2011). The Digital Mistrust composite demonstrated acceptable internal consistency ($\alpha = .79$, 95% CI [.78, .91]), indicating stable and reliable measure of user mistrust. Similarly, the digital privacy concerns composite exhibited acceptable reliability ($\alpha = .78$, 95% CI [.77, .79]).

Exploratory factor analysis provided additional evidence for construct validity. A one factor solution for Digital Mistrust composite demonstrated a strong model fit, with high factor loadings across items ($\lambda = .78-.83$), accounting for 53% of the total variance. Parallel analysis conducted using 'fa.parallel()' also supported retaining one underlying factor, as only the first observed exceeded the reference threshold (see Figure 1.). In contrast, the digital privacy concerns composite exhibited a weaker one-factor structure, explaining 48% of the variance with acceptable loadings ($\lambda = .53-.82$). Given these results, the Privacy Concerns composite was retained for predictive and exploratory purposes, with results interpreted more cautiously. Together these findings reinforce the reliability of mistrust measure.

Figure 1. Parallel analysis supporting a one factor solution for the Digital Mistrust composite.



Descriptive statistics indicated that digital privacy concerns and digital mistrust were both meaningfully present across the sample, with standardized composite scores centered near zero but exhibited substantial variability. The standardized digital mistrust composite ranged from -2.55 to 1.27, with a median of 0.09 and an interquartile range from -0.49 to 0.69. This distribution suggests that while most respondents clustered near the sample average, a meaningful subset exists and exhibits elevated levels of mistrust (Figure 2). Similarly, the standardized privacy concerns composite ranged from -2.41 to 1.27, with a median of 0.04 and an interquartile range from -0.51 to 0.6 (Figure 3.). When looked at together, these distributions indicate that privacy concerns and mistrust are widespread but heterogenous, which reinforces the need for specified data-driven approached rather than a uniform, single digital policy response.

Exploratory visualizations further revealed some of the relationships between digital mistrust and privacy concerns. A bivariate scatterplot indicated that a positive association between digital privacy concerns corresponded to higher mistrust levels (Figure 4.). In contrast, privacy literacy showed negative association with mistrust (Figure 5.), suggesting that informed users and clearer policies may mitigate distrust rather than exacerbate it. These exploratory patterns motivated the formal hypothesis tests presented further.

Applied probability showed the statistical patters and translated them into interpretable quantities for decision making. A binary indicator of high digital mistrust was defined as a standardized score above the sample mean. Using this classification, the probability that a randomly selected adult exhibited high digital mistrust was 0.54 indicating that more than half the respondents approach digital platforms with elevated mistrust. Furthermore, given the near-universal adoption of smartphones in the sample ($P(\text{Owns smartphone}) = 0.974$), conditional probabilities were examined to asses whether digital access mitigates any digital mistrust. The probability of high mistrust among owners was 0.546, nearly identical to the overall rate, which suggests that widespread technology adoption does not correspond to lower mistrust. This finding underscores a central business challenge which is that increased platform reach does not guarantee user confidence or trust. Additionally, privacy literacy emerged as a meaningful differentiator variable. Among individuals with high privacy literacy, the probability of high digital mistrust was 0.425 compared to 0.594 for users with lower literacy. This substantial difference indicates that users who better understand privacy regulations and data practices are significantly less likely to exhibit elevated mistrust. This may suggest that user education and clarity might reduce disengagement driven by uncertainty.

Behavioral responses to privacy threats further illustrated the defensive nature of mistrust. Individuals who actively declined online tracking exhibited a higher probability of high mistrust at 0.566 than those who did not at 0.480. Rather than reflecting trust, these behaviors might signal heightened concern and caution toward the digital systems they use.

Finally, privacy concerns strongly differentiated mistrust levels. Among individuals with above-average privacy concerns, the probability of high digital mistrust was 0.614 compared to 0.466 among those with low concerns. This gap highlights that privacy concerns may be a central driver of mistrust.

Figure 2. Distribution of the Digital Mistrust Composite

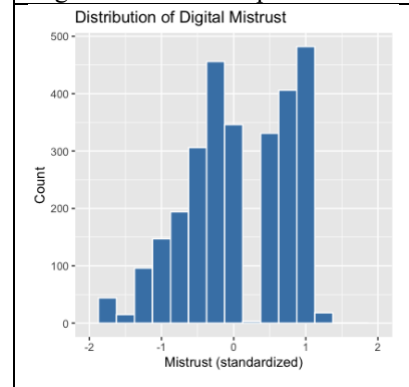
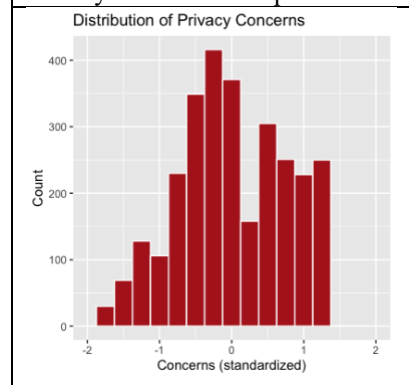


Figure 3. Distribution of the Privacy Concerns Composite



Correlation analysis revealed several meaningful associations aligned with the study's hypotheses. Digital mistrust was positively correlated with digital privacy concerns ($r = .16$), indicating that individuals who express greater concern about data practices also tend to report higher mistrust. Perceived effectiveness of privacy policies

Table 4. Linear Regression Model Predicting Digital Mistrust	
Variable	Model 1 (p-value)
Intercept	-0.355* (0.013)
Privacy Concerns	0.130* (< .001)
Privacy Literacy	-0.226* (< .001)
Policy Effectiveness	0.251* (< .001)
Declines Tracking	-0.099* (0.002)
Uses Encrypted Apps	0.056* (0.053)
Uses Private Browser	0.004 (0.888)
Changes Social Media Settings	-0.102* (.002)
Age (categories)	ns
Education: Some College	0.229* (0.006)
Education: High School	0.161 (0.058)
Income	0.004 (0.457)
Internet Frequency	-0.014 (0.528)
Gender	ns
Moderate Republican	-0.144* (<.001)
Moderate Liberal	-0.244*(<.001)
Liberal	-0.086*(0.016)
Political Ideology	-0.036 (0.002)
Smartphone Ownership	0.001 (0.994)
R ²	0.223
Adjusted R ²	0.217

exhibited the strongest association with digital mistrust ($r = .37$), suggesting that clearer and more effective policies are strongly linked to higher mistrust. Privacy literacy was moderately negatively correlated with mistrust ($r = -.27$), reinforcing the idea that informed users experience less uncertainty and disengagement (Table. 3)

In contrast, demographic variables showed weak or negligible correlations with digital mistrust, indicating that mistrust is more closely tied to perceptions and experiences than to age, education, or income alone. Digital habit variables were modestly intercorrelated with one another, reflecting clusters of privacy-protective behaviors, but showed weaker direct associations with mistrust. Overall, these correlations support subsequent regression analyses by demonstrating that privacy perceptions are more salient predictors of mistrust than basic demographics.

Multiple linear regression was used to test the hypotheses regarding the predictors of digital mistrust while controlling for digital behaviors and demographics. The full model was statistically significant ($F(23, 2894) = 36.17, p < .001$) and explained approximately 20.7% of the variance in digital mistrust ($R^2 = .22$; adjusted $R^2 = .22$), indicating meaningful explanatory power for the psychological and behavioral outcomes. Consistent with hypothesis 1, digital privacy concerns were a significant positive predictor of digital mistrust ($\beta = 0.13, SE = 0.02, p < 0.01$). Individuals who expressed higher concerns reported higher levels of mistrust, even after accounting for literacy, behaviors and demographics. This finding reinforces the central business problem of privacy concerns directly undermining the user's trust in digital systems. Hypothesis 2 was also supported as privacy literacy exhibited a strong negative association with digital mistrust ($\beta = -0.23, SE = 0.02, p < 0.01$), indicating that more informed users were substantially less mistrustful. From a business intelligence perspective, this suggests that clarity, education and transparency may mitigate mistrust rather than exacerbate it. Regarding results related to perceived effectiveness, there was statistical significance, but unexpected directionally. In contrast to hypothesis 3, higher perceived policy effectiveness was associated with higher digital mistrust ($\beta = 0.25, SE = 0.02, p < .001$). This suggests that users who closely focus on privacy policies may do so because of underlying concern of skepticism, further highlighting the complex relationship of transparency and trust. Digital habits demonstrated mixed effects. Declining online tracking ($\beta = -0.10, p < .002$) and changing social media privacy settings ($\beta = -0.11, p < .001$) were associated with lower mistrust, while the use of encrypted messaging applications showed a modest positive association with mistrust ($\beta = 0.06, p = .053$). Taken together, these findings support Hypothesis 4, indicating that digital behaviors jointly contribute to mistrust, though individual habits vary

in direction and magnitude. Demographic effects were more nuanced. Compared to respondents with less than a high school education, individuals with some college education reported higher levels of mistrust ($\beta = 0.23, p =$

.006), while other education categories did not differ significantly. Political ideology emerged as a meaningful predictor, with several ideological groups exhibiting significantly lower mistrust relative to the conservative reference category. Age, income, internet frequency, gender, and smartphone ownership were not statistically significant once psychological perceptions and digital behaviors were included, suggesting that mistrust is driven primarily by experiential and attitudinal factors rather than basic demographic characteristics (Table 4).

Ten-fold cross validation was used to evaluate the stability of linear regression model predicting digital mistrust. The cross validated model produced an RMSE of 0.7, an R^2 of 0.2 and a mean absolute error of 0.55 which closely matched the in-sample model performance (Table 5.). These results indicate that the model generalizes well across data partitions and that the observed relationships were not driven by overfitting. From a business intelligence perspective, this suggests that the model can reliably inform decisions related to trust building and privacy strategy across similar user populations.

To continue, the analysis of variance indicated that digital mistrust did not significantly differ across age groups, $F(3, 2914) = 0.13, p = .94$ or gender categories ($F(2, 2915) = 1.37, p = .26$). In contrast, educational attainment was a significant source of variation in digital mistrust ($F(5, 2912) = 4.78, p < .001$), indicating that mistrust levels differ systematically by education level. These results suggest that demographic segmentation based on education may be more informative for addressing digital mistrust than age or gender alone, reinforcing the importance of informational and literacy-based interventions.

Logistic regression was used to predict the probability of high digital mistrust. Privacy concerns significantly increased the odds of high mistrust ($OR = 1.52, p < .001$), while privacy literacy significantly reduced those odds ($OR = 0.58, p < .001$). Perceived privacy policy effectiveness was also positively associated with high mistrust ($OR = 1.95, p < .001$), replicating the counterintuitive pattern observed in the linear model (Table 3.).

Several digital behaviors were significant predictors. Declining online tracking ($OR = 0.80, p = .020$) and changing social media privacy settings ($OR = 0.71, p < .001$) were associated with lower odds of high mistrust, whereas other behaviors and most demographic variables were not significant. Education remained a significant protective factor ($OR = 0.89, p < .001$). Overall, the results indicate that elevated digital mistrust is primarily driven by privacy perceptions and knowledge rather than demographic characteristics.

IV. Conclusion

The trust and transparency business intelligence project examined how privacy perceptions, knowledge, and digital behaviors shape public trust in technology platforms. Across descriptive, probabilistic, and inferential analyses the findings all converged on a clear conclusion: digital mistrust is widespread, predictable and strongly tied to how organizations design, communicate, and implement data practices. More than half of respondents showed an above average digital mistrust, underscoring that digital mistrust is not a marginal sentiment but a structural challenge for modern digital organizations.

From a business performance standpoint, the results point to several actionable insights. Privacy concerns consistently proved that they were a primary driver of mistrust while privacy literacy acted as a protective factor. These findings suggest that investments done by companies and government entities should focus on user education, clarity and accessible explanations of data practices that may reduce mistrust more effectively than policy expansion alone. Additionally, creating less invasive privacy policies that respect the interests of the user are important as they are the foundation of what create digital mistrust in the first place. At the same time, the counter intuitive finding that perceived policy effectiveness was associated with higher mistrust highlights a critical point of friction, which is that transparency without usability or credibility may heighten skepticism rather than alleviate it. Organizations should therefore prioritize clear, meaningful and user-centered privacy communication, rather than solely on policy compliance.

Furthermore, behavioral findings indicate that many privacy protective actions such as declining tracking, adjusting social media privacy settings, using private browsers or using encrypted messaging apps are likely reactive responses to mistrust and not indicators of confidence. These behaviors reflect and reinforce the importance of addressing mistrust upstream rather than downstream, before users disengage and adopt behaviors that reduce data quality and platform effectiveness. Maintaining the same privacy policies that create defensive behaviors in users also creates the risk of increased psychological consequences in user's lives further separating the individuals from the digital platforms that organizations want interaction with. Education based segmentation also appears more informative than demographic targeting by age or gender which suggests that knowledge-based intervention may outperform broad demographic strategies. It is also worth noting that as seen in Figure 3 there may be different groups of people that have different needs when it comes to digital mistrust so using a blanket strategy for all users may not be most efficient. More user-focused research needs to be conducted to understand the differences between user groups.

Several limitations were also encountered during the project and require acknowledgment. First, the analysis relies on self-reported survey data which may be subject to response bias and cannot establish causal relationships. Second, while regression models explain a meaningful portion of variance in digital mistrust, a substantial share remains unaccounted for which reflects the complexity of understanding the psychological trust process. Third, the use of cross-sectional data limits the ability to assess how trust may evolve over time. Fourth, due to the nature of needing to use an omission strategy for data management as compared to a mean imputation strategy omitted many responses around ~2000. While the data set is still large, omission needed to occur due to the composite score creation. Despite these limitations, the selected analytical methods were all justified. Linear regression captured variation in mistrust levels, logistic regression enabled probability-based risk classification, and cross-validation demonstrated that results were stable and generalizable rather than artifacts of overfitting.

From a broader perspective, this project demonstrates how business intelligence can bridge technical data analysis and human-centered decision making. Trust is not merely an outcome, but through the project a measurable, data-driven construct with direct implications for engagement, retention and system adoption. By integrating psychometric validation, probability reasoning, and predictive modeling, this analysis provides organizations with a framework for evaluating trust risks and evaluating the downstream effect that privacy practices can have. Ultimately, this BI project can inform more ethical, transparent and effective digital systems, hopefully paving the way for platforms that not only collect data more efficiently and ethically but also earn and sustain the trust of the public.

Appendix A.

Figures.

Figure 1. Parallel analysis supporting a one factor solution for the Digital Mistrust composite.

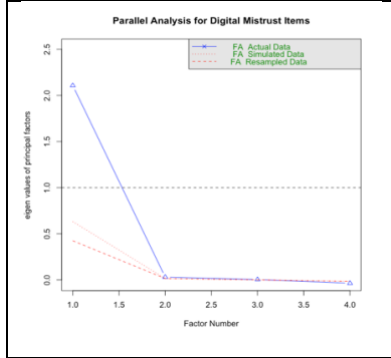


Figure 2. Distribution of the Digital Mistrust Composite

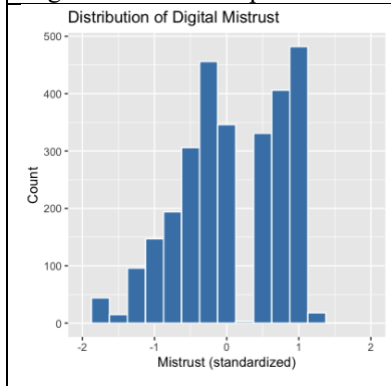


Figure 3. Distribution of the Privacy Concerns Composite

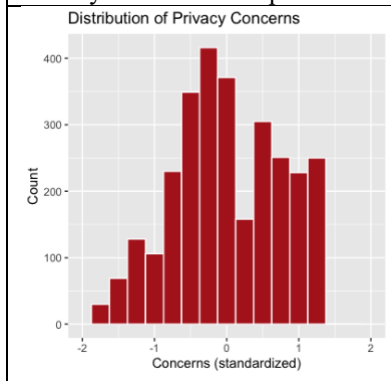


Figure 4. Digital Privacy Concerns Composite vs Digital Mistrust Composite

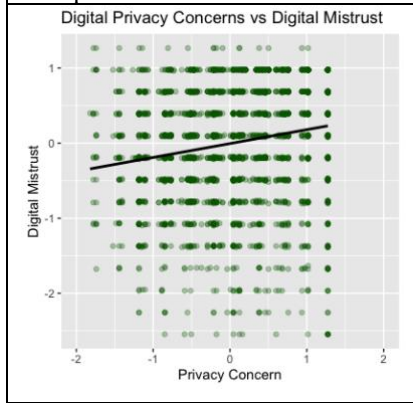
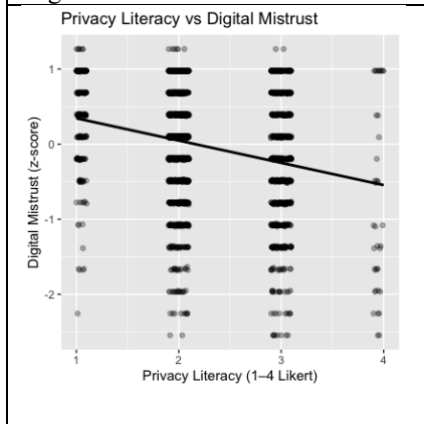


Figure 5. Perceived Effectiveness vs Digital Mistrust



Appendix B.

Tables

Table 1. Conceptual Pathway from Data Practices to Digital Engagement		
Organizational Data Practices	User Psychological Response	Digital Platform Outcomes
Data collection methods Privacy policy design Transparency level	Perceived control Privacy concern Trust	Platform engagement System adoption Willingness to share data

Table 2. Hypothesis Testing Plan				
#	Null Hypothesis	Alt. Hypothesis	H ₀ (symbolic)	H _a (symbolic)
1	Privacy concerns do not predict digital mistrust	Higher privacy concerns predict higher digital mistrust	$\beta_{\text{conc}} = 0$	$\beta_{\text{conc}} > 0$
2	Privacy literacy does not predict digital mistrust.	Greater privacy literacy predicts lower digital mistrust.	$\beta_{\text{priv_lit}} = 0$	$\beta_{\text{priv_lit}} < 0$
3	Perceived privacy policy effectiveness does not predict digital mistrust.	Higher perceived policy effectiveness predicts lower digital mistrust.	$\beta_{\text{ppeff}} = 0$	$\beta_{\text{ppeff}} < 0$
4	Digital habits have no joint effect on mistrust	Digital habits have a joint effect on mistrust.	$\beta_{\text{track}} = \beta_{\text{encrypt}} = \beta_{\text{browser}} = \beta_{\text{social}} = 0$	At least one $\beta \neq 0$
5	Digital mistrust does not differ by age group	Digital mistrust differs by age group.	$\mu_{18-29} = \mu_{30-49} = \mu_{50-64} = \mu_{65+}$	At least one group mean differs

Table 3. Correlation Matrix of Key Variables												
Variable	trust_z	conc_z	ppeff	priv_lit	track	encrypt	browser	social	age	edu	income	int_freq
trust_z	1.00	0.16	0.37	-0.27	-0.10	0.04	-0.01	-0.06	0.01	0.01	0.07	-0.02
conc_z	0.16	1.00	0.08	0.02	-0.16	-0.15	-0.14	-0.12	0.04	-0.06	-0.05	-0.05
ppeff	0.37	0.08	1.00	-0.22	-0.07	0.02	-0.00	-0.05	-0.05	0.17	0.20	-0.08
priv_lit	-0.27	0.02	-0.22	1.00	-0.08	-0.18	-0.14	-0.13	-0.01	-0.01	-0.04	-0.03
track	-0.10	-0.16	-0.07	-0.08	1.00	0.20	0.24	0.25	-0.04	-0.04	-0.03	-0.01

encrypt	0.04	-0.15	0.02	-0.18	0.20	1.00	0.29	0.27	0.11	-0.08	-0.03	0.11
browser	-0.01	-0.14	-0.00	-0.14	0.24	0.29	1.00	0.26	0.09	-0.05	-0.04	0.06
social	-0.06	-0.12	-0.05	-0.13	0.25	0.27	0.26	1.00	0.24	-0.11	-0.05	0.14
age	0.01	0.04	-0.05	-0.01	-0.04	0.11	0.09	0.24	1.00	-0.08	0.02	0.31
edu	0.01	-0.06	0.17	-0.01	-0.04	-0.08	-0.05	-0.11	-0.08	1.00	0.40	-0.21
income	0.07	-0.05	0.20	-0.04	-0.03	-0.03	-0.04	-0.05	0.02	0.40	1.00	-0.13
int_freq	-0.02	-0.05	-0.08	-0.03	-0.01	0.11	0.06	0.14	0.31	-0.21	-0.13	1.00

Table 4. Linear Regression Model Predicting Digital Mistrust	
Variable	Model 1 (p-value)
Intercept	−0.355* (0.013)
Privacy Concerns	0.130* (< .001)
Privacy Literacy	−0.226* (< .001)
Policy Effectiveness	0.251* (< .001)
Declines Tracking	−0.099* (0.002)
Uses Encrypted Apps	0.056* (0.053)
Uses Private Browser	0.004 (0.888)
Changes Social Media Settings	−0.102* (.002)
Age (categories)	ns
Education: Some College	0.229* (0.006)
Education: High School	0161 (0.058)
Income	0.004 (0.457)
Internet Frequency	−0.014 (0.528)
Gender	ns
Moderate Republican	−0.144* (<.001)
Moderate Liberal	−0.244*(<.001)
Liberal	−0.086*(0.016)
Political Ideology	−0.036 (0.002)
Smartphone Ownership	0.001 (0.994)
R ²	0.223
Adjusted R ²	0.217
F-statistic (p-value)	54.17 (< .001)

Table 5. Logistic Regression Predicting High Digital Mistrust	
Variable	Odds Ratio
Privacy Concerns (conc_z)	1.52*
Privacy Literacy (priv_lit)	0.58*
Policy Effectiveness (ppeff)	1.95*
Declines Tracking	0.80*
Uses Encrypted Apps	1.17
Uses Private Browser	1.07
Changes Social Media Settings	0.71*
Education	0.89*
Other controls	n.s.
AIC	3573.4

Table 6. Cross-Validation Performance Metrics (10-Fold CV)	
Metric	Value
RMSE	0.70
R ²	0.20
MAE	0.55
Folds	10
Sample Size	2,918

Appendix C

References

- Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behavior in the age of information. *Science*, 347(6221), 509–514. <https://doi.org/10.1126/science.aaa1465>
- Büchi, M., Fosch-Villaronga, E., & Tamò-Larrieux, A. (2021). The chilling effects of algorithmic profiling: Mapping the issues. *Computer Law & Security Review*, 40, 105559.
- Dinev, T., & Hart, P. (2006). An extended privacy calculus model for e-commerce transactions. *Information Systems Research*, 17(1), 61–80. <https://doi.org/10.1287/isre.1060.0080>
- Reynolds, C., Altmann, R., & Allen, D. (2021). *Mastering modern psychological testing: Theory and methods* (2nd Edition). Springer: Cham, Switzerland.
- Pew Research Center. (2019). Americans and privacy: Concerned, confused and feeling lack of control over their personal information. <https://www.pewresearch.org/internet/2019/11/15/americans-and-privacy/>
- Pew Research Center. (2023). How Americans view data privacy. <https://www.pewresearch.org/internet/2023/10/18/how-americans-view-data-privacy/>
- Stoycheff, E., Liu, J., Wibowo, K. A., & Nanni, D. P. (2017). Privacy and the panopticon revisited: Online surveillance and the chilling effect on political expression. *Journal of Communication*, 68(2), 318–342. <https://doi.org/10.1093/joc/jqx001>
- Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, 2, 53–55. <https://doi.org/10.5116/ijme.4dfb.8dfd>

Appendix D

R Functions

Data Import and Preparation

`rm(list = ls())` – Clear workspace

`read_sav()` (haven) – Import SPSS dataset

`library()` – Load required packages

`names()` – Rename variables

`as.numeric()` – Convert labeled variables to numeric

`factor()` – Recode categorical variables

`reverse_code()` – Custom function for reverse-coding Likert items

Logical indexing (`x[x == value] <- NA, %in%`) – Recode invalid responses

Missing Data Handling and Dataset Construction

`is.na()` – Identify missing values

`table(..., useNA = "ifany")` – Inspect missing/invalid codes

`complete.cases()` – Complete-case filtering

Subsetting (`data[, ]`) – Create master analytic dataset

Composite Score Construction

`scale()` – Standardize items (z-scores)

`rowMeans()` – Compute composite scores

Reliability and Construct Validation

`psych::alpha()` – Cronbach's alpha

`fa.parallel()` (psych) – Parallel analysis

`fa()` (psych) – Exploratory factor analysis

Descriptive Statistics and Correlations

`summary()` – Descriptive statistics

`cor()` – Pearson correlation matrix

`round()` – Output formatting

Visualization

`ggplot()` (ggplot2) – Plot framework

`geom_histogram()`, `geom_density()` – Distributions

`geom_point()`, `geom_smooth(method = "lm")` – Scatterplots

`geom_boxplot()` – Group comparisons

`scale_x_continuous()`, `labs()`, `theme()` – Plot formatting

Applied Probability

`ifelse()` – Binary indicators

`mean()` – Probability estimation

Logical operators (`&`, `|`) – Joint and conditional probabilities

`median()` – Variable dichotomization

Inferential Modeling

`lm()` – Multiple linear regression

`anova()` – Joint significance testing

`confint()` – Confidence intervals

`aov()` – ANOVA

`glm(family = binomial)` – Logistic regression

`exp()` – Odds ratios

`predict(type = "response")` – Predicted probabilities

`table()` – Confusion matrix

Model Validation

`caret::train()` – k-fold cross-validation

`trainControl()` – Resampling specification